Higher National Diploma in Engineering (Civil)

Second Year, Second Semester Examination - 2021

CE 2212-Structural Design

Q1)

A rectangular beam of 250mm x 600mm is simply supported with the effective span of 8.0m. It carries a dead load of 20kN/m with the self-weight and imposed load of 15kN/m. Grade of concrete, reinforced steel and shear links are 30N/mm, 460N/mm and 250N/mm respectively.Design the reinforcement for bending and shear with the relevant checks97. Consider the cover to the reinforcement as 20mm.

(Total 20 marks)

Q2)

A reinforced concrete continuous beam sized 250mm x 800mm has three equal spans of 6m. It carries a 200mm thick RC slab spanned 4m centers in transverse direction. This flanged beam carries a total dead load of 80kN/m with the self-weight and imposed load of 60kN/m. Grade of concrete, reinforced steel and shear links are 30N/mm^2, 460N/mm^2 and 250N/mm^2 respectively.Design the bending and shear reinforcements for the middle of end span and the 1st interior support. Carry out the relevant checks for the beam. Consider the cover to the reinforcement as 20mm.

(Total 20 marks) [107]

Q3)

Four equal spans reinforced concrete slab spanning in one direction (Fig 01) carries an imposed load of 2.5kN/m^2. The effective span is 6.0m. Finishes and ceiling load of the slab is1.5KN/m^2. Grade of concrete and reinforced steel are 30N/mm^2 and 460N/mm^2 respectively.Consider the cover to the reinforcement as 20mm. Design reinforcements for the maximum moment. Carry out the relevant checks for the slab (Shear & deflection).

(Total 20 marks)

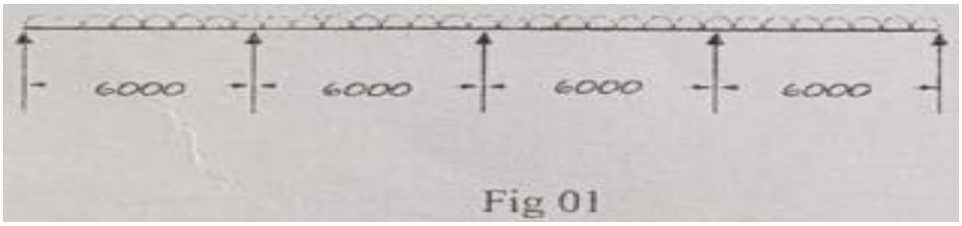

Fig 01

Q4)

A single timber column **200mm x 200mm** is required to carry a concentrated point load of **210kN**. The load has been transferred to the column by timber joists such that the end restraint conditions top and bottom may be taken as restrained in **position but not in direction**. The height of column is **2.8m** and the timber may be taken as strength class **C27**. The load may be considered as **short term** and the member is used in **wet exposure condition**. Check the suitability of the given section for the above load.

(Total 20 marks)

Q5)

a) A lap joint consists of two steel plates connected by **six 12mm diameter bolts of grade 4.6**. If the plates are **8mm thick (S275)** and stretched by **40kN force** as shown in the fig 2. Edge and end distances are **50mm**. Check suitability of the connection for the given force. (10 marks)

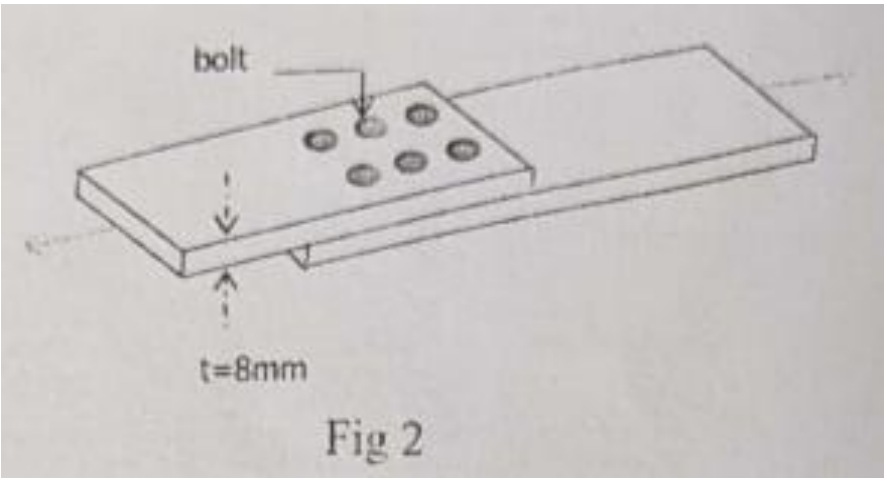

Fig 2

b) A steel bracket is bolted to the column flange as shown in fig 3. The steel is Grade S275 and thickness is less than 16mm. Check the adequacy of the bolts if the bolt diameter is 20 mm in grade 8.8. (10 marks)

(Total 20 marks)

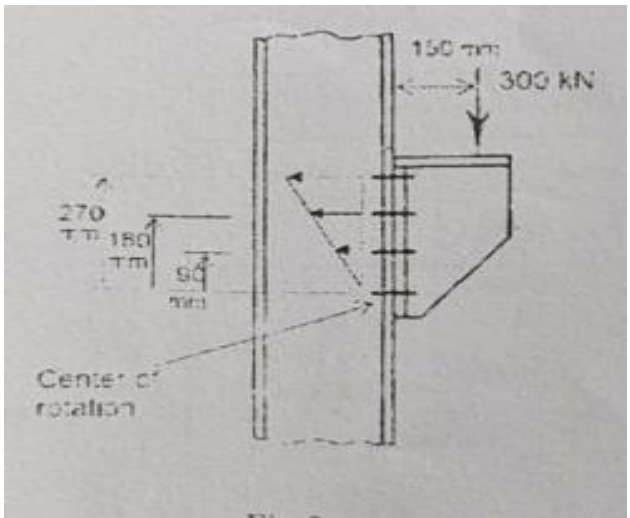

Fig 3

Q6)

The internal load-bearing brick wall shown in fig 4, supports an ultimate axial load of **140 kN/m run** including self-weight of the wall[140]. The wall is **102.5 mm thick** and **4m long**. Assuming the masonry units conform to **Category II** and the construction control category is **'normal'**, design the wall.

(Total 20 marks)

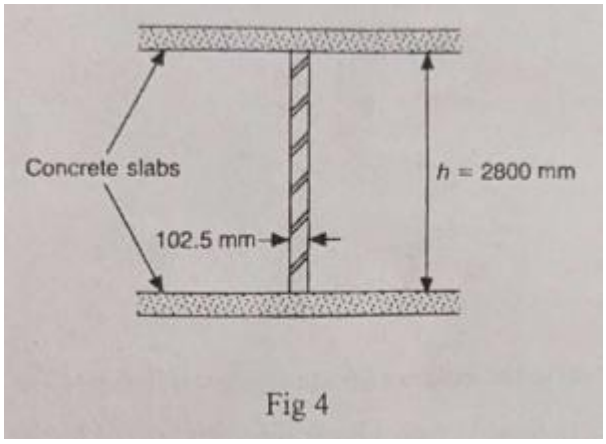

Fig 4

Table 32. Strength of bolts in clearance holes

	Bolt grade		Other grades of fasteners
	gr 4.6	gr 8.8	
	N/mm²	N/mm²	N/mm²
Shear strength, p_s	160	375	$0.48U_f$ but $\leqslant 0.69\ Y_f$
Bearing strength p_{bb} (but see table 33)	460	1035	$0.72\ (U_f + Y_f)$
Tension strength, p_t	195	450	$0.58U_f$ but $\leqslant 0.83\ Y_f$

Y_f is the specified minimum yield strength of the fastener.

U_f is the specified minimum ultimate tensile strength of the fastener.

Table 33. Bearing strength of connected parts for ordinary bolts in clearance holes, p_{bs}

Steel to BS 4360			Other grades of steel
gr 43	gr 50	gr 55	
N/mm²	N/mm²	N/mm²	N/mm²
460	550	650	$0.65(U_s + Y_s)$

Y_s is the specified minimum yield strength.

U_s is the specified minimum ultimate tensile strength.

Bearing Strength of Connected plates

Steel Grade	S275	S355	S460
Bearing Strength(N/mm^2)	460	550	670

Shear capacity of the bolt, $P_s = A_s p_s$

Bearing Capacity of the bolt, $P_{bb} = d t_b p_{bb}$

Bearing capacity of connected plates, $P_{bs} = k_{bs} d t_p p_{bs}$ but $P_{bs} \leq 0.5 k_{bs} e t_p p_{bs}$

Tension Capacity of the bolt, $P_t = A_t p_t$

A_s = Shear area

k_{bs} = 1 for Standard clearance holes

e = End distance

d = Nominal diameter of bolt

t_p = Thickness of the connected plates

A_t = Tensile stress area

p_t = Tension strength

Tensile stress area

Nominal size and tread diameter (mm)	Tensile stress area (mm^2)
12	84.5
16	157
20	245
22	303
24	353

6.3.6.3 *Combined shear and tension.* When bolts are subject to both shear and tension then in addition to the conditions in 6.3.1 to 6.3.6.2 the following relationship should be satisfied:

$$\frac{F_s}{P_s} + \frac{F_t}{P_t} \leq 1.4$$

where

F_s is the applied shear;

F_t is the applied tension;

P_s is the shear capacity (see 6.3.2);

P_t is the tension capacity (see 6.3.6.1).

Area's of Reinforcement

Cross-sectional areas of groups of bars (mm^2)

Bar Size	Number of Bars * Denotes non-preferred sizes									
	1	2	3	4	5	6	7	8	9	10
6mm	28.3	56.5	84.8	113	141	170	198	226	255	283
8mm	50.3	100	151	201	251	302	352	402	452	503
10mm	78.5	157	236	314	339	471	550	628	707	785
12mm	113.1	226	339	452	566	679	792	905	1718	1131
16mm	201.1	402	603	804	1005	1206	1407	1608	1810	2011
20mm	314.2	628	942	1257	1571	1885	2199	2513	2827	3142
25mm	490.9	982	1473	1963	2454	2945	3436	3927	4418	4909
32mm	804.2	1608	2413	3217	4021	4825	5630	6434	7238	8042
40mm	1257.0	2513	3770	5026	6283	7540	8790	10053	11310	12566
50mm*	1963.5	3927	5890	7854	9812	11781	13744	15708	17671	19635

Cross-sectional areas for Specific Spacings in mm^2 / Metre Width

Bar Size	Spacing of Bars (Millimetres) * Denotes non-preferred sizes									
	75	100	125	150	175	200	225	250	275	300
6mm *	377	283	226	188	162	141	125	113	103	94
8mm	670	503	402	335	287	251	223	201	183	168
10mm	1047	786	628	524	449	393	348	314	285	262
12mm	1508	1131	905	754	646	565	503	452	411	377
16mm	2681	2011	1608	1340	1149	1005	894	801	731	670
20mm	4189	3142	2513	2094	1795	1571	1396	1257	1143	1047
25mm	6545	4909	3927	3272	2805	2454	2182	1963	1785	1636
32mm	-	8042	6434	5362	4596	4021	3574	3217	2924	2681
40mm	-	-	10053	8378	7181	6283	5587	5027	4571	4189
50mm*	-	-	-	13090	11220	9617	8727	7854	7140	6545

Table 4a) — Partial safety factors for material strength — Compression, γ_m

		Category of construction control	
		Special	Normal
Category of manufacturing control of structural units	Special	2.5	3.1
	Normal	2.8	3.5

Table 7 — Capacity reduction factor, β

Slenderness ratio h_{ef}/t_{ef}	Eccentricity at top of wall, e_x			
	Up to $0.05t$ (see note 1)	$0.1t$	$0.2t$	$0.3t$
0	1.00	0.88	0.66	0.44
6	1.00	0.88	0.66	0.44
8	1.00	0.88	0.66	0.44
10	0.97	0.88	0.66	0.44
12	0.93	0.87	0.66	0.44
14	0.89	0.83	0.66	0.44
16	0.83	0.77	0.64	0.44
18	0.77	0.70	0.57	0.44
20	0.70	0.64	0.51	0.37
22	0.62	0.56	0.43	0.30
24	0.53	0.47	0.34	
26	0.45	0.38		
27	0.40	0.33		

NOTE 1 It is not necessary to consider the effects of eccentricities up to and including $0.05t$.

NOTE 2 Linear interpolation between eccentricities and slenderness ratios is permitted.

NOTE 3 The derivation of β is given in Appendix B.

28.3.1 *Effective height*

28.3.1.1 *Walls*

The effective height of a wall may be taken as:

a) 0.75 times the clear distance between lateral supports which provide enhanced resistance to lateral movement, or

b) the clear distance between lateral supports which provide simple resistance to lateral movement.

23.1.1 ***Walls or columns of small plan area***

Where the horizontal cross-sectional area of a loaded wall or column is less than 0.2 m², the characteristic compressive strength should be multiplied by the factor:

$(0.70 + 1.5A)$

where

A is the horizontal loaded cross-sectional area of the wall or column (m²).

23.1.2 ***Narrow brick walls***

When brick walls are constructed so that the thickness of the wall or loaded inner leaf of a cavity wall is equal to the width of a standard format brick, the values of f_k obtained from Table 2a) may be multiplied by 1.15.

28.3.2 *Effective length*

The effective length of a wall may be taken as:

a) 0.75 times the clear distance between vertical lateral supports or twice the distance between a support and a free edge, where lateral supports provide enhanced resistance to lateral movement;

b) the clear distance between lateral supports or 2.5 times the distance between a support and a free edge where lateral supports provide simple resistance to lateral movement.

28.4 Effective thickness

The effective thickness of a wall, column or pier is given in **28.4.1** and **28.4.2** and is illustrated in Figure 3.

28.4.1 *Walls and columns not stiffened by piers or intersecting walls*

For single leaf walls and columns the effective thickness is the actual thickness.

For cavity walls and columns the effective thickness should be taken as two-thirds the sum of the actual thicknesses of the two leaves or the actual thickness of the thicker leaf, whichever is the greater.

28.4.2 *Walls stiffened by piers or intersecting walls*

Where a wall, which may be one leaf of a cavity wall, is stiffened by piers, the effective thickness, t_{ef}, of the wall, or leaf of a cavity wall, is:

32.2 Design strength of masonry

The design strength of masonry is the characteristic strength multiplied by a capacity reduction factor, β, and divided by the appropriate partial safety factor for material.

32.2.1 *Design vertical load resistance of walls*

The design vertical load resistance of a wall per unit length is given by:

$$\frac{\beta t f_k}{\gamma_m}$$

where

β is a capacity reduction factor allowing for the effects of slenderness and eccentricity and is obtained from Table 7;

f_k is the characteristic strength of the masonry obtained from clause 23;

γ_m is the appropriate partial safety factor for the material obtained from clause 27 Table 4a);

t is the thickness of the wall.

Table 2 — Characteristic compressive strength of masonry, f_k, in N/mm²

a) Constructed with standard format bricks

Mortar designation	Compressive strength of unit (N/mm²)								
	5	10	15	20	27.5	35	50	70	100
(i)	2.5	4.4	6.0	7.4	9.2	11.4	15.0	19.2	2[illegible].0
(ii)	2.5	4.2	5.3	6.4	7.9	9.4	12.2	15.1	18.2
(iii)	2.5	4.1	5.0	5.8	7.1	8.5	10.6	13.1	15.5
(iv)	2.2	3.5	4.4	5.2	6.2	7.3	9.0	10.8	12.7